

Hướng dẫn build firmware Xiaozhi AI Tiếng Việt

1. Cài đặt môi trường

- Git: <https://git-scm.com/install/windows>

- Visual Studio Code: <https://code.visualstudio.com/>

- Python: (Python 3.10 is the minimum supported version for ESP-IDF.)

<https://www.python.org/downloads/windows/>

- EIM chọn theo máy windows hay macbook

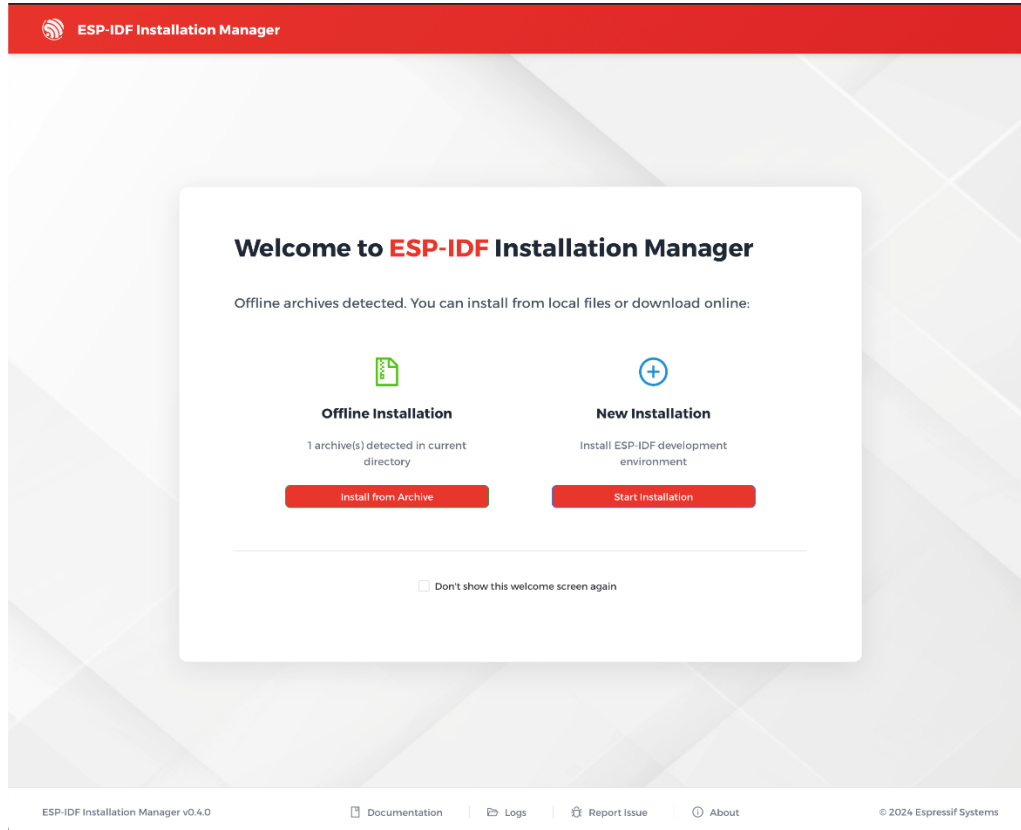
<https://dl.espressif.com/dl/eim/>

- ESP-IDF v6.1 chọn theo máy windows hay macbook

<https://dl.espressif.com/dl/eim/?tab=offline>

- Cài đặt:

GUI-Based Offline Installation



1. Mở ứng dụng cài đặt (EIM).
2. Nếu file ``.zst`` nằm cùng thư mục với EIM thì sẽ được tự động phát hiện.
Nếu không: bấm ****New Installation**** → ****Offline Installation**** hoặc ****Browse Archive File****.
3. Chọn file lưu trữ ``.zst``.
4. Màn hình tiếp theo hiển thị file đã chọn, cho phép cấu hình ****Installation Path**** (dùng đường dẫn mặc định hoặc tự chọn).
5. Bấm ****Start Installation**** để bắt đầu. Quá trình cài hiển thị tương tự bản expert.



Offline Installation

[< Back](#)

Installation Configuration

Selected Archive



archive_v5.5.zst



Installation Path

/Users/petrgadorek/espresif

[Browse](#)

☒ Use default installation path

Cancel

Start Installation



Offline Installation

[< Back](#)

Installing ESP-IDF from Offline Archive

Current Activity:

Extracting archive: archive_v5.5.zst

Processing offline archive contents

Overall Progress

20%

1

Check

Validating archives

2

Extract

Extracting archive contents

3

Prerequisites

Installing dependencies

4

Install

Installing ESP-IDF

5

Tools

Setting up development tools

6

Python

Configuring Python environment

7

Configure

Finalizing configuration

8

Complete

Installation complete

[Installation Log >](#)

(3 entries)

2. Lấy mã nguồn

```
git clone https://github.com/78/xiaozhi-esp32
cd xiaozhi-esp32
git submodule update --init --recursive
```

3. Tùy chỉnh code (tùy chọn)

- Mở thư mục bằng VS Code để chỉnh sửa code, thêm các chức năng:
 - Thông báo tin tức
 - Giá vàng
 - Tích hợp MCP / Home Assistant...
- Các phần này nằm trong main/, cấu hình ở main/Kconfig.

4. Build với ESP-IDF

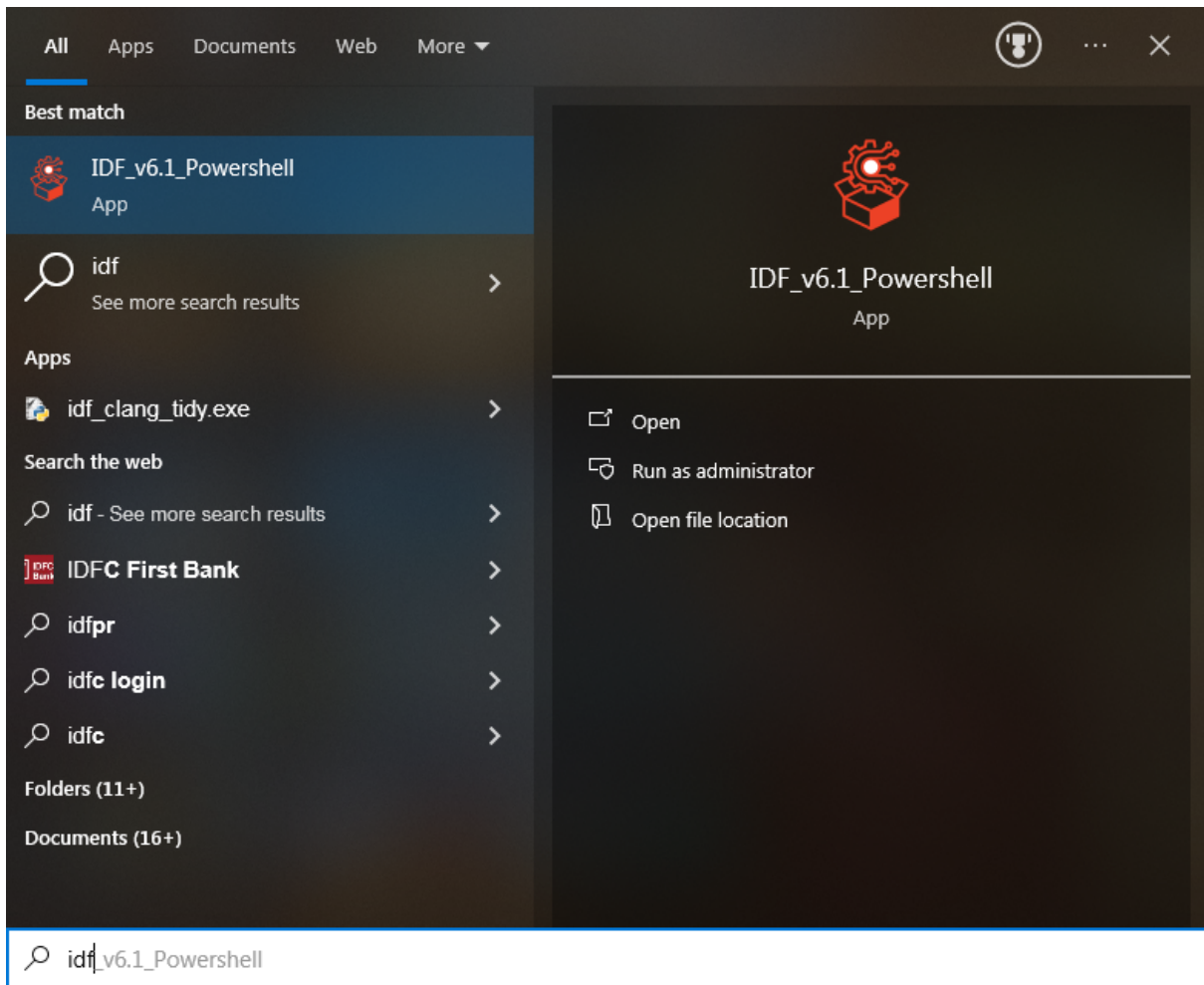
Mở terminal **IDF CMD / IDF PowerShell v6.1**, rồi:

```
cd xiaozhi-esp32  
idf.py set-target esp32s3  
idf.py menuconfig
```

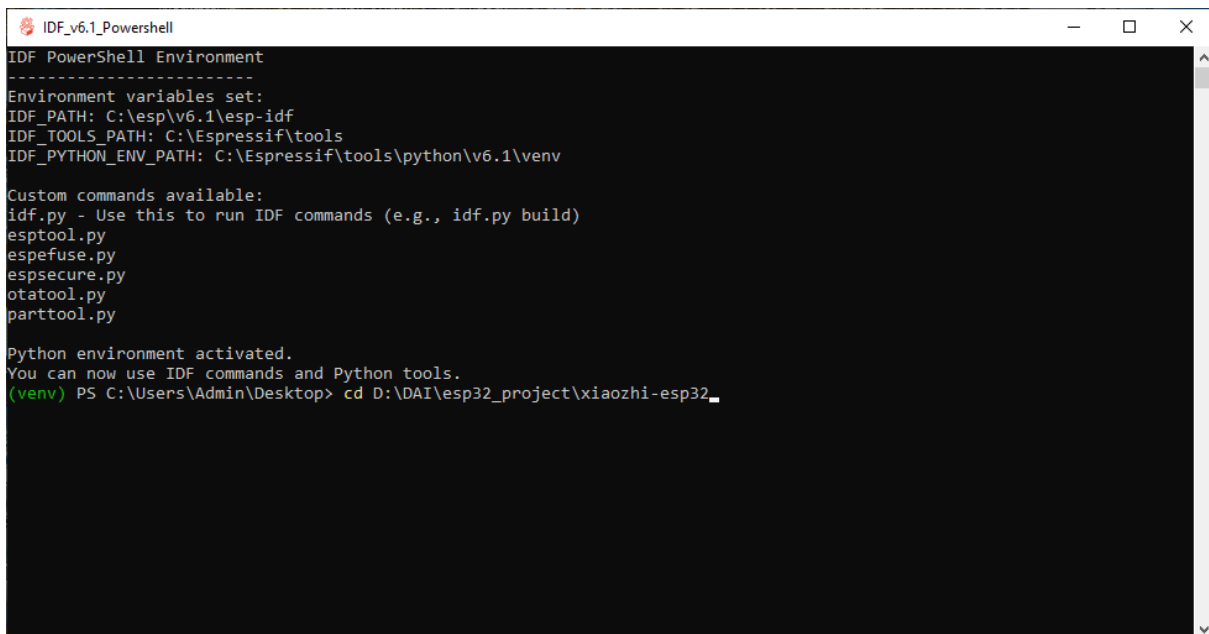
Trong menuconfig:

- **Xiaozhi Assistant** → **Board Type** → chọn board của bạn, theo dõi các mục:
 - Video: chọn loại board (VD: M5StickC Plus, board tự làm...)
 - Chọn **ngôn ngữ**
 - Chọn **loại màn hình**
 - Cấu hình loa / mic
 - Cấu hình server ASR nhận diện giọng nói tiếng Việt (VD FunASR / Whisper) tại **Audio** hoặc **Network/ASR**

Mở IDF_v6.1_Powershell



cd đến thư mục xiaozhi_esp32



code . để mở VS Code

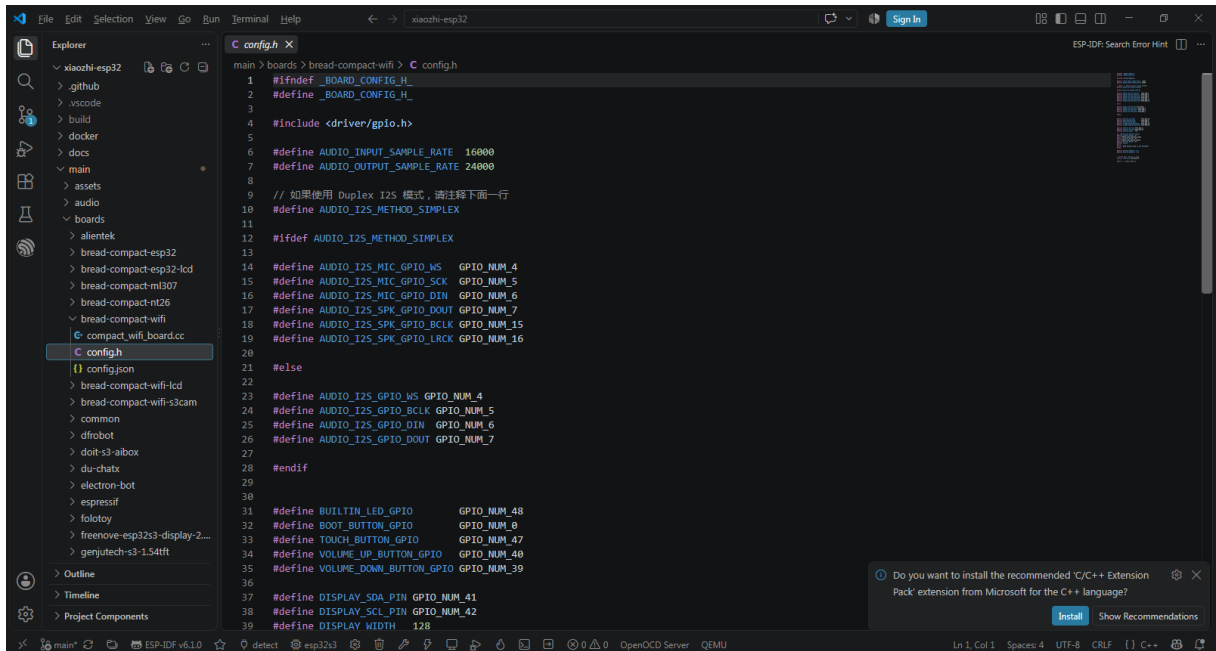
```
IDF_v6.1_Powershell
IDF PowerShell Environment
-----
Environment variables set:
IDF_PATH: C:\esp\v6.1\esp-idf
IDF_TOOLS_PATH: C:\Espressif\tools
IDF_PYTHON_ENV_PATH: C:\Espressif\tools\python\v6.1\venv

Custom commands available:
idf.py - Use this to run IDF commands (e.g., idf.py build)
esptool.py
espefuse.py
espesecure.py
otatool.py
parttool.py

Python environment activated.
You can now use IDF commands and Python tools.
(venv) PS C:\Users\Admin\Desktop> cd D:\DAI\esp32_project\xiaozhi-esp32
(venv) PS D:\DAI\esp32_project\xiaozhi-esp32> code .
```

Sửa dòng 38 để ngôn ngữ mặc định cho chatbot là Tiếng Việt. Xem cấu hình của board ở thư mục tên board tương ứng

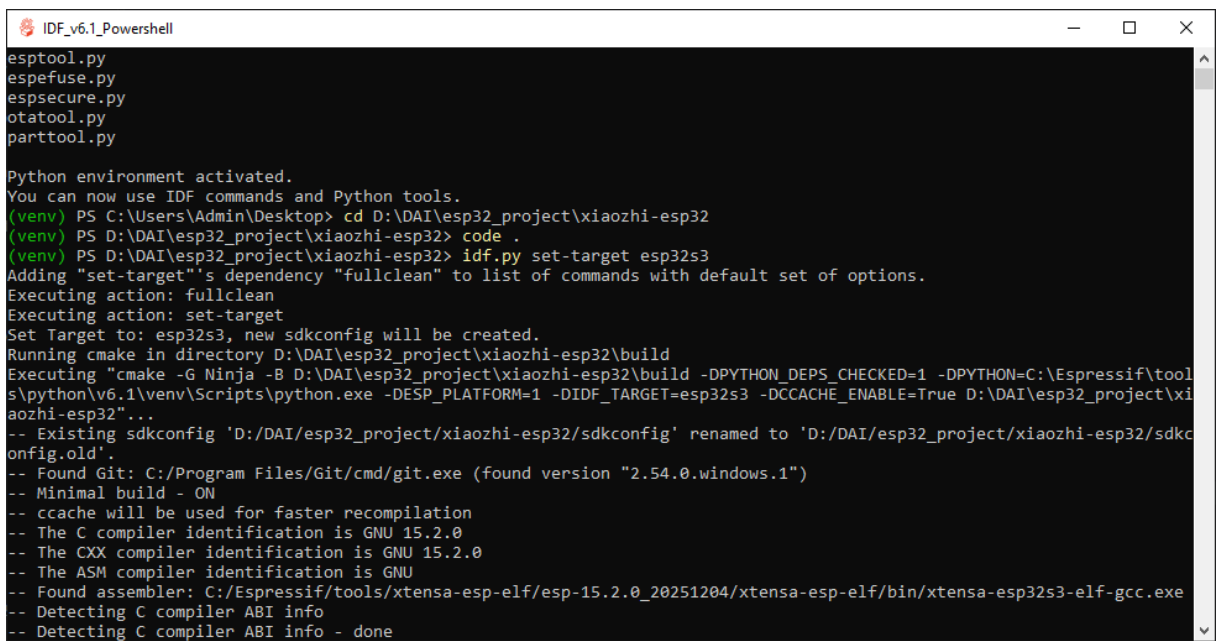
```
main > Kconfig.projbuild
9 choice
10 prompt "Flash Assets"
11 help
12   Select the assets to flash.
13
14 config FLASH_NONE_ASSETS
15   bool "Do not flash assets"
16 config FLASH_DEFAULT_ASSETS
17   bool "Flash Default Assets"
18   depends on !USE_EMOTE_MESSAGE_STYLE
19 config FLASH_CUSTOM_ASSETS
20   bool "Flash Custom Assets"
21 config FLASH_EXPRESSION_ASSETS
22   bool "Flash Emote Assets"
23   depends on USE_EMOTE_MESSAGE_STYLE
24 endchoice
25
26 config CUSTOM_ASSETS_FILE
27   depends on FLASH_CUSTOM_ASSETS
28   string "Custom Assets File"
29   default "assets.bin"
30   help
31     The custom assets file to flash.
32     It can be a local file relative to the project directory or a remote url.
33 choice
34   prompt "Default Language"
35   default LANGUAGE_VI_VN
36   help
37     Select device display language
38
39 config LANGUAGE_ZH_CN
40   bool "Chinese"
41 config LANGUAGE_ZH_TW
42   bool "Chinese Traditional"
43 config LANGUAGE_EN_US
44   bool "English"
45 config LANGUAGE_JA_JP
46   bool "Japanese"
```



Cách thêm 1 board chưa có sẽ hướng dẫn sau.

Ok rồi quay lại IDF_v6.1_Powershell dùng các lệnh sau để build và flash:

```
idf.py set-target esp32s3
idf.py menuconfig
idf.py build
idf.py build flash monitor
idf.py merge-bin
```



```
IDF_v6.1_Powershell
32/managed_components/espressif_usb_host_uvc C:/esp/v6.1/esp-idf/components/esptool_py C:/esp/v6.1/esp-idf/components/f
atfs C:/esp/v6.1/esp-idf/components/freertos C:/esp/v6.1/esp-idf/components/hal C:/esp/v6.1/esp-idf/components/heap C:/e
sp/v6.1/esp-idf/components/http_parser D:/DAI/esp32_project/xiaozhi-esp32/managed_components/laride_heatshrink C:/esp/v
6.1/esp-idf/components/log D:/DAI/esp32_project/xiaozhi-esp32/managed_components/lvgl_lvgl C:/esp/v6.1/esp-idf/componen
ts/lwip D:/DAI/esp32_project/xiaozhi-esp32/managed_components/m5stack_m5ioe1 D:/DAI/esp32_project/xiaozhi-esp32/managed
_components/m5stack_m5pm1 D:/DAI/esp32_project/xiaozhi-esp32/main C:/esp/v6.1/esp-idf/components/mbedtls C:/esp/v6.1/es
p-idf/components/nvs_flash C:/esp/v6.1/esp-idf/components/nvs_sec_provider C:/esp/v6.1/esp-idf/components/partition_tabl
e C:/esp/v6.1/esp-idf/components/pthread C:/esp/v6.1/esp-idf/components/sdmmc C:/esp/v6.1/esp-idf/components/soc C:/esp/
v6.1/esp-idf/components/spi_flash C:/esp/v6.1/esp-idf/components/spiffs C:/esp/v6.1/esp-idf/components/tcp_transport D:/
DAI/esp32_project/xiaozhi-esp32/managed_components/tny-robotics_sh1106-esp-idf D:/DAI/esp32_project/xiaozhi-esp32/manag
ed_components/txp666_otto-emoji-gif-component C:/esp/v6.1/esp-idf/components/vfs D:/DAI/esp32_project/xiaozhi-esp32/man
aged_components/waveshare_custom_io_expander_ch32v003 D:/DAI/esp32_project/xiaozhi-esp32/managed_components/waveshare_
esp_lcd_sh8601 D:/DAI/esp32_project/xiaozhi-esp32/managed_components/waveshare_esp_lcd_touch_cst9217 C:/esp/v6.1/esp-id
f/components/wear_levelling C:/esp/v6.1/esp-idf/components/wpa_supplicant D:/DAI/esp32_project/xiaozhi-esp32/managed_com
ponents/wvirgil1123_sscca_client C:/esp/v6.1/esp-idf/components/xtensa
-- Configuring done (60.5s)
-- Generating done (1.9s)
-- Build files have been written to: D:/DAI/esp32_project/xiaozhi-esp32/build
(venv) PS D:\DAI\esp32_project\xiaozhi-esp32> idf.py menuconfig
```

```
IDF_v6.1_Powershell
(Top)
Build type --->
App Update config ----
Bootloader config --->
Security features --->
Application manager --->
Boot ROM Behavior --->
Serial flasher config --->
Partition Table --->
Xiaozhi Assistant --->
Expression Emote Configuration --->
ESP Speech Recognition --->
Compiler options --->
Component config --->
[ ] Make experimental features visible (default value)

q Quit s Save o Load d Save min / Search ? Info f Help c Name a All r Reset p Palette
```

Xiaozhi Assistant ->

- Chọn Target Borad (chọn board đang lập trình)
- Chọn OLED Type (chọn màn hình, một số board không có mục này)

```
IDF_v6.1_Powershell
(Top) > Xiaozhi Assistant
<-- Back
(https://api.tenclass.net/xiaozhi/ota/) Default OTA URL (default value)
Flash Assets (Flash Default Assets) (default value) --->
Default Language (Vietnamese) (default value) --->
Target Board (Bread Compact Wi-Fi (ESP32)) (default value) --->
OLED Type (SSD1306 128*64) --->
Select display style (Enable default message style) (default value) --->
[ ] Use multiline chat message display (default mode only) (default value)
Wake Word Implementation Type (Wakenet model with AFE) (default value) --->
[*] Send Wake Word Data (default value)
[ ] Enable Wake Word Detection in Listening Mode (default value)
[*] Enable AFE Audio Processing (default value)
[ ] Enable Server-Side AEC (Unstable) (default value)
[ ] Enable Audio Debugger (default value)
WiFi Configuration Method --->
[ ] Enable Custom Message Reception (default value)
Camera Configuration --->

q Quit s Save o Load d Save min / Search ? Info f Help c Name a All r Reset p Palette
```

Tùy chọn: từ đánh thức Hi Jason

```
IDF_v6.1_Powershell
(Top) > ESP Speech Recognition > Load Multiple Wake Words (WakeNet9s)
<-- Back
[ ] Hi, (wn9s_hilexin) (default value)
[ ] Hi,ESP (wn9s_hiesp) (default value)
[ ] (wn9s_nihaoxiaozhi) (default value)
[*] Hi,Jason (wn9s_hijason)
```

Bấm q => Y để save. Sau đó idf.py build

5. Biên dịch

```
idf.py build
```

```
IDF_v6.1_Powershell
Loading defaults file D:/DAI/esp32_project/xiaozhi-esp32/sdkconfig.defaults...
NOTE: C:\Users\Admin\AppData\Local\Temp\kconfgen_tmpstz2p6fd:33: CONFIG_NEWLIB_NANO_FORMAT was replaced with CONFIG_LIBC_NEWLIB_NANO_FORMAT
Loading defaults file D:/DAI/esp32_project/xiaozhi-esp32/sdkconfig.defaults.esp32s3...
NOTE: Kconfig parser version: 1
NOTE: Kconfig defaults policy: use sdkconfig
NOTE: Status: Finished with notifications
NOTE: C:/esp/v6.1/esp-idf/components/esp_libc/Kconfig:107: LIBC_NEWLIB_NANO_FORMAT has user-set value y from C:\Users\Admin\AppData\Local\Temp\kconfgen_tmpstz2p6fd but is not visible (disabled dependency)
NOTE: D:/DAI/esp32_project/xiaozhi-esp32/managed_components/lvgl_lvgl/Kconfig:1229: LV_USE_CALENDAR_HEADER_ARROW has user-set value n from C:\Users\Admin\AppData\Local\Temp\kconfgen_tmpstz2p6fd but is not visible (disabled dependency)
NOTE: D:/DAI/esp32_project/xiaozhi-esp32/managed_components/lvgl_lvgl/Kconfig:1233: LV_USE_CALENDAR_HEADER_DROPDOWN has user-set value n from C:\Users\Admin\AppData\Local\Temp\kconfgen_tmpstz2p6fd but is not visible (disabled dependency)
HINT: Disabled symbols/choices are not written into sdkconfig file. Setting their user value has no effect.
NOTE: Loaded configuration 'D:/DAI/esp32_project/xiaozhi-esp32/sdkconfig'
Configuration saved to 'D:/DAI/esp32_project/xiaozhi-esp32/sdkconfig'

(venv) PS D:\DAI\esp32_project\xiaozhi-esp32> idf.py build
```

Build xong thì flash. Đã kết nối thiết bị với cổng USB của máy tính.

6. Nạp firmware và xem log

```
idf.py flash monitor
```

(Thoát monitor bằng `Ctrl + ]`)

7. Gộp firmware (tùy chọn)

```
idf.py merge-bin
```

File gộp nằm trong build/ (VD merged.bin), dùng để flash qua web ESP Home Flasher hoặc tool ngoài.